

**ISE 441:  
Operations Research and Optimization Methods  
Fall 2026**

Instructor: Dr. Ramin Moghaddas  
Phone: 305- 284-9505  
Email: [ramin@miami.edu](mailto:ramin@miami.edu)  
[UM Webpage](#)

In-Person Class Hours and Location: Tuesdays and Thursdays:  
**Tu, Th 11:00 AM - 12:15 PM**  
**Dooly Memorial 213**

Office Hours:  
**1:00 PM – 2:00 PM on Tuesdays & Thursdays**  
and **anytime** by appointment

\*\* Please note that all policies in this syllabus will be strictly followed. However, exceptions may be made on a case-by-case basis for students who are unable to attend class due to Q/I status or because of delayed arrival.

“All students are bound by the [Academic Integrity Policy](#), which is intended to promote and support an atmosphere conducive to learning. Each student’s work is to be their own unless express permission is given by the instructor. Additionally, while students are bound by the policy, faculty are also bound to report violations as offered in the policy.”

**Important Dates:**

- **Test 1:** Sept 29
- **Test 2:** Nov 10
- **Final Exam (Comprehensive):** Check Canelink (Dec 3-9)
- **Project 1 Report:** Dec 9, 11 PM
- **Group Project Presentation (Dec 1)**

## 1. Overview

*“Operations Research (OR), or operational research in the U.K, is a discipline that deals with the application of advanced analytical methods to help make better decisions. The terms management science and analytics are sometimes used as synonyms for operations research”* [1]. While Operations Research is known as one of the most important topics in industrial Engineering, it is also widely applicable in Computer Engineering, Computer Science, and many other disciplines (e.g. biostatistics, business, and management). This class is an introduction to the fundamental methods used in deterministic operations research and optimization. Topics covered will include linear programming, solution methods, transportation models, network flows, integer programming, and dynamic programming. We will discuss both modeling and theory with some (occasional) proofs. We will work with basic software packages to have experience in computer applications.

## 2. Course Objectives

By the end of this course, students are expected to

- Understand the importance and broad applicability of OR
- Model deterministic OR problems
- Understand the basic mathematical theory of OR and solution methods
- Understand the basics of the most commonly used optimization algorithms
- Learn the basics of Computer modeling and programming in OR problems
- 

## 3. Course Materials

Primary and Required Text: Hamdy A. Taha, *Operations Research: An Introduction*, Prentice Hall. 9<sup>th</sup> Edition. (ISBN 10: 0-13-255593-X)

- 10<sup>th</sup> Edition is also available, but we focus only on the 9<sup>th</sup> edition.

Suggested Text: Wayne L. Winston, *Operations Research: Applications and Algorithms*, Duxbury. (ISBN: 0-534-38058-1)

*Note: Instructor will post additional materials on Canvas for interested readers throughout the semester.*

## 4. Course Outline

- Introduction to Optimization
- Modeling with Linear Programming (LP)
- Software Packages for Linear Programming (LP)
- The Simplex Method
- Duality and Sensitivity Analysis

---

1. <https://www.informs.org>

- Transportation Models & Network models
- Integer Programming (IP)
- Multi-Objective Optimization & Goal Programming (GP)
- Dynamic Programming (DP)

## 5. Other Course Materials

- This course is lecture based, primarily using slides in PDF format, which will be made available in **Canvas**. Only the raw version of each lecture slide (in PDF) will be posted on the Canvas. It is your responsibility to take notes in class (or online), if needed. The instructor will not provide any annotated version for lecture notes.
- All other supplementary materials (Frequently Asked Questions (FAQ), a set of examples used in class, solution guide for assignments, sample midterms, case studies, a discussion of other topics in Operations Research & a reading list for interested students, and other relevant materials) will be posted in Canvas throughout the term.

## 6. Access to the Instructor

- You are welcome to **contact the instructor anytime by email. The title of all emails should be IEN 441** to receive a quick reply.
- If you have questions about fairness in grading an assignment (please check solution manuals first), or an issue or suggestion about the course, you can contact the instructor.
- You can always ask for an appointment with the instructor through email. Emails to the instructor will be answered by a maximum **48 hours**. Answers to important questions will be posted on the course **FAQ**, for the benefit of all.

## 7. Marking Scheme

Assignments: 15%

Attendance & Quizzes \*: 10% (total of 6 small quizzes with or without prior announcement, out of which only 5 are considered).

Test # 1: 15%

Test # 2: 15%

Final Exam: 20%

Project: 25% + bonus only for **outstanding** work (up to 5%)

*\* You may miss the whole 10% for poor attendance.*

## 8. Assignments and Projects

- Assignments are to be completed individually. Any noticeable similarity between your submission and that of another student will be considered as cheating.
- This course has one group project. Detailed information for the project and expected deliverables will be provided in class. If the instructor rejects your proposal, you have 5 days to resubmit. If it gets rejected again, then you will have to work on a specific topic that will be chosen by the instructor.
- There is no specific format for assignment submissions: typed and handwritten are both acceptable. Make sure they are clear and readable.

- The project report has a specific format.
- Make sure that you put your **full name and student number** on your assignment so that you get credit for the work.
- Review the solution guide, and only contact the instructor if there is indeed an issue.
- Assignments are to be done individually and without help.

### 9. Late to Class and Poor Attendance

- When students come to class late, it can disrupt the flow of the lecture and distract the instructor and other students. Students who show up late for more than 2-3 lectures or on a regular basis (without prior notice to instructor) will most likely lose the whole 10% for attendance/quiz.
- Students who miss the lectures on a regular basis will most likely lose the entire 10% for attendance/quiz.

### 10. Late Assignment

Assignments are due on Canvas by the midnight of the due date and **cannot be accepted** after the deadline (no exceptions).

### 11. Late Project

Late project (final report) will only be accepted due to a serious medical or personal emergency. A project that is otherwise late will have a penalty as follows:

- 50% penalty within the first 24 hours of being late
- The project will not be accepted after 24 hours.

### 12. Tests

- The tests are not strictly cumulative. However, the contents of the exams are not mutually exclusive and tend to build on previous material, so concepts introduced in one part of the course often reappear later in the course.
- All exams will be held *in class* (60-75 minutes in duration) and **will not be open notes. Final exam may be longer!**

### 13. Re-exam Rule

If a re-exam is granted (under very strict circumstances), **the exam will be comprehensive** (i.e., use material from the entire course as covered in that lecture section).

### 14. Course Feedback

In this course, the instructor welcomes constructive feedback (positive or negative) and any suggestions for improvements to the course throughout the term. Feedback (at least 200 words) can also be submitted to get up to 1% extra credit. Course feedback must be handed in by **Dec 1st** (the mechanism for submitting your feedback will be announced later). Feedback can be submitted in a sealed envelope.

### 15. Other Important Course Rules (PLEASE READ)

- Please check Canvas frequently.

- All overheads, notes, and electronic materials for the course are subject to copyright. Reproduction for distribution other than for your personal use is prohibited unless explicit specific permission is granted.

**- Recording of lectures is prohibited.**

- Students are encouraged to ask their questions during the lecture.

- Students are encouraged to attend each lecture, as most exam questions will be directly or indirectly discussed in class. ***It is your responsibility to get the notes from a classmate if you miss a lecture.*** Students are expected to attend lectures to succeed in this course.

- Students are **not allowed to use cell phones** during the class.

- Laptops, notebooks, pads, or any electronic devices (except for cell phones) are allowed during lectures **only** if it is for taking notes (Texting, web surfing, emailing, chatting, etc. are not allowed).

- The information contained in the syllabus may be subject to change with reasonable advance notice, as deemed appropriate by the instructor.

**- Cheating and/or plagiarism will not be tolerated.** If you are caught on a quiz, exam or homework, you will fail that quiz, exam or homework. If this becomes a repeated offense, you will be referred to the appropriate individuals. **Academic Integrity and Honor Code will be strictly observed and enforced!!!**

**- Religious Holidays** – The University of Miami, although a secular institution, is determined to accommodate those students who wish to observe religious holy days. It seeks to reflect its awareness of and sensitivity to religious holy days whenever possible when scheduling University activities. The following provisions are meant to apply equitably to all religious groups and to provide opportunities to all to meet their religious obligations.

Except as specifically provided to the contrary, this policy is binding on all students in undergraduate programs. Schools offering graduate or professional programs, including undergraduate professional programs, are strongly encouraged to adhere to these policies to the maximum extent practicable.

Any student absent from class in observance of a religious holy day shall not be penalized in any way for an examination or assignment missed during the period of absence. Absence in observance of a religious holy day does not relieve students from responsibility for any part of the course work required during the period of absence. Students who are absent on days of examinations or class assignments shall be offered a reasonable opportunity to make up the work without penalty, if the student previously arranged to be absent. Nothing in this policy shall preclude faculty members from limiting the number of student absences to a reasonable number of absences for any reason. The faculty member has discretion to determine how the make-up obligation will be fulfilled. A faculty member who penalizes a student contrary to these provisions may have committed unprofessional conduct, and thus may be subject to a complaint to the Committee on Professional Conduct under the provisions of Section B4.9 of the Faculty Manual.

It is the student's obligation to provide faculty members with notice of the dates they will be absent due to observance of religious holy days, preferably before the beginning of classes but no later than the end of the first three class days. For religious holy days that fall within the first three class days, students must provide faculty members with notice no later than two class days before the absence. Missing a class due to travel plans associated with a particular religious holy day does not constitute an excused absence. Absences due to observance of religious holy days that are not pre-

arranged with the relevant faculty member within the first three class days may be considered unexcused, and the faculty member may therefore prevent the student from making up examinations or assignments missed during the period of absence.

Faculty members are encouraged to anticipate days when a substantial number of students will be absent for observance of religious holy days and should avoid scheduling examinations and assignment deadlines on those days. Faculty members are expected to reasonably assist students in obtaining class information the student missed during the period of absence in observance of a religious holy day. In that regard, faculty members are urged to allow taping or recording of the class session, with the reproduction limited to the student's personal use, when a student misses a class due to observance of a religious holy day. To assist in identifying religious observance days, faculty members are encouraged to consult the illustrative list provided in the Interfaith Calendar. Faculty members are urged to remind students of their obligation to inform faculty members within the first three class days of any anticipated absences due to observance of religious holy days and should include that information in the syllabus or course requirements document for that course.

For more information you can refer to the University Bulletin - <https://petal.miami.edu/teaching-guides-and-policies/religiousholiday/>

- Make sure that you log on at least a few minutes before the scheduled class time so that we can start on time.
- All tests and presentations will be during the regular course time.
- All assignments should be submitted through Assignment Tool in Canvas. Students can use Adobe Scanner on their device (phone or iPad) to scan documents into PDFs. The application is free for students with the UM Adobe licensing.
- You can always request more office hours and the instructor will do his best to set up sessions upon your request.
- For each important topic in the course, the instructor will create a Canvas Discussion Board. Using the Canvas Discussion Board, the instructor can create a forum where students can create a post and respond to their peers with text, files, images, video, and audio. You need to check Canvas frequently for all updates.
- All grades will be submitted through Canvas.
- Remote students (with prior approval from the university) should adhere to the same standards of behavior during the video class session that you would follow in a real classroom setting. Be punctual and courteous. Be professional in your communication with participants. Be mindful of all video class participants. Do not interrupt while someone else is speaking. Allow other participants time and opportunities to contribute to the discussion and share their ideas with the group.

- Be mindful of your tone and expressions. Speak clearly and avoid shouting. Video class sessions are not anonymous. Your voice and video are viewed by all who are participating. Sessions will be recorded and saved for students currently in different time zones.
- Be forgiving of mistakes during the video class session. There are bound to be technical glitches; be patient with the moderator as well as the other participants.
- Students who are attending a class session synchronously are required to have their video enabled.
- Students are expressly prohibited from recording any part of this course. Meetings of this course might be recorded by the University. Any recordings will be available to students registered for this class as they are intended to supplement the classroom experience. Students are expected to follow appropriate University policies and maintain the security of passwords used to access recorded lectures. Recordings may not be reproduced, shared with those not in the class, or uploaded to other online environments. If the instructor or a University of Miami office plans any other uses for the recordings, beyond this class, students identifiable in the recordings will be notified to request consent prior to such use.
- If you are approved to take this course under the Remote Learning Option, attendance in the virtual class is required as scheduled unless this creates undue hardship due to differences in your residential time-zone and that of Miami Florida. If you are a Remote Learning Option student, you may not under any circumstances physically attend the class on campus. If you cannot attend the virtual class due to illness or other reason, you must contact the instructor. Unexcused absences from the classroom may affect your grade or lead to failing the course.
- If you are approved to take this course under the Remote Learning Option, you must keep up with the virtual class as scheduled. You may not under any circumstances physically attend the class on campus. If you cannot keep up with the virtual class due to illness or other reason, you must contact the instructor. Failure to keep up with the virtual class as scheduled may affect your grade or lead to failing the course.
- Professor Ramin Moghaddass is the copyright owner of the courseware; individual recordings of the materials on Canvas and/or of the virtual sessions are not allowed; and that such materials cannot be shared outside the physical or virtual classroom environment.

### ***Diversity & Inclusion Statement***

The College of Engineering embraces a notion of an intellectual community enriched and enhanced by diversity along a number of dimensions, including race, ethnicity and national origins, gender and gender identity, sexuality, class, and religion. In order to create a learning environment for my students that supports a diversity of thoughts, perspectives, and experiences:

- If you have a name and/or set of pronouns that differ from those that appear in your official UM records, please let me know!
- If you feel like your performance in the class is being impacted by your experiences outside of class, please don't hesitate to come and talk with me. I want to be a resource for you. You can also notify me of the issue through another source such as your academic advisor, a trusted faculty

member, or a peer. If for any reason you do not feel comfortable discussing the issue directly with me, I encourage you to seek out another, more comfortable avenue to address the issue.

- I (like many people) am still in the process of learning about diverse perspectives and identities. If something was said in class (by anyone) that made you feel uncomfortable, please talk to me about it.

### **Generative Artificial Intelligence (AI)**

Software may be useful tools for enhancing learning, productivity, and creativity. For instance, they can assist with brainstorming, finding information, and creating materials, such as text, images, and other media. However, these tools must be used appropriately and ethically, and you must understand their limitations. In particular, it is important to realize that all AI software has the following limitations:

- How output is arrived at is not clear as the internal processes used to produce a particular output within the generative AI cannot be determined.
- AI output is typically based on data harvested from unknown online sources. As such, it may reflect biases that should be acknowledged. AI output may also be inaccurate or entirely fabricated, even if it appears reliable or factual.
- AI evokes a range of intellectual property concerns; sourcing and ownership of information is often unclear and is currently the subject of ongoing litigation.
- If you use AI tools in any part of your work, you are responsible for the final product of that work, both academically and in the workforce.
- If you use AI in the project, you need to clearly cite the tool and the section you used AI for. Using AI for assignment and Project (as a solution tool) is not allowed.